NetMind: Review 2 checklist

AI-Powered Network Intelligence and LLM Network Copilot

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Repository	github.com/mo-affan/NetMind
Prepared	15 October 2026

How to use this: go through the checklist on page 1 the evening before and again 30 minutes before the review. Sections 3 to 8 are what you present; section 9 is what to show live; section 10 is the viva preparation. Every number in this document is real output from NetMind or from our NS2 run; where data is synthetic, it says so, and you should say so too.

1. Before the review: checklist

Laptop

- ☐ **git pull** in C:\Users\moham\NetMind, then **python -m pip install -r requirements.txt** (Streamlit 1.51+ for dark mode)
- ☐ **python -m pytest -q** passes (note the pass count to quote it)
- ☐ **python -m netmind demo** runs and shows 3 incidents
- ☐ **python -m netmind dashboard** opens at http://localhost:8501 (keep it running in a second window)
- ☐ Dashboard: a **Synthetic demo** run and the **real NS2 trace** (ns2\results\netmind.tr) are both loaded as runs, so you can switch between them without waiting
- ☐ Copilot answers "Why is the network slow?" on the Overview page
- ☐ Ubuntu: **~/netmind_ns2** has the NS2 output (netmind.tr, netmind.nam); **python3 check_trace.py netmind.tr** shows 5/5 PASS
- ☐ NAM opens (export TCL_LIBRARY / TK_LIBRARY if needed); if not, screenshots or the dashboard instead
- ☐ Charger, screen zoom set so terminal text is readable, other apps closed, notifications off

Material

- ☐ Review 2 slides (architecture, modules, results, NS2, demo, next steps)
- ☐ Review 1 proposal and the faculty feedback (NS2 added) to show it was acted on
- ☐ Screenshots as backup: dashboard Overview, Incidents, Topology with a failed link, Copilot, terminal demo, NS2 check output, NAM at 3 s / 7 s / 8.5 s
- ☐ Who presents which part is agreed (section 3)

2. The 30-second opening

> “Network monitoring tools show numbers like latency and packet loss but leave it to the user to work out what is wrong. NetMind measures the network from its traffic, detects unusual behaviour with an Isolation Forest model, and explains it in plain language through a Copilot that combines the measurements with a networking knowledge base. We use three kinds of input: real captures, an NS2 simulation with congestion and a link failure at known times, and a synthetic demo with three known problems, so we can check that what NetMind finds matches the truth.”

Problem and objectives

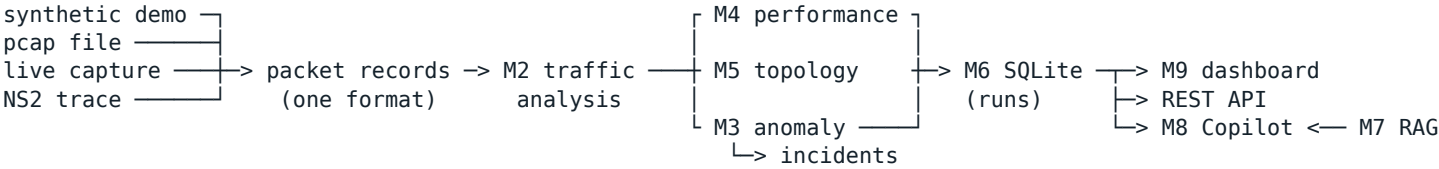
- **Problem:** raw metrics (RTT, loss, connection counts) need expertise to interpret; problems such as congestion, scans or floods are noticed late.
- **Objective 1, measure:** protocols, flows, latency, throughput, loss and jitter from traffic.
- **Objective 2, detect:** flag unusual time windows and group them into typed incidents.
- **Objective 3, explain:** a Copilot that answers questions using the run's data and retrieved networking knowledge (RAG), with sources. Advisory only; it never changes the network.
- **Objective 4, routing:** topology graph, Dijkstra, Link State vs Distance Vector, link failure.
- **Objective 5, simulate:** NS2 scenario with congestion and rerouting to validate the analysis.

3. Who presents what

Part	Modules	Presenter
Input and traffic analysis	M1 capture (Scapy, pcap, synthetic), M2 protocols and flows	Hardik
Topology and routing, storage	M5 graph, Dijkstra, LS, DV; M6 SQLite	Hardik
Dashboard	M9 Streamlit dashboard (8 pages)	Hardik (lead)
Performance and anomaly detection	M4 metrics; M3 Isolation Forest, deviation check, incidents	Affan
Knowledge and Copilot	M7 RAG; M8 Copilot (offline, Claude, Ollama)	Affan
NS2 simulation	TCL scenario, trace parser, checker, lab run	Affan

Adjust if you split it differently; the owners match the README.

4. Architecture



- **Key design rule:** every input becomes the same packet-record table (time, addresses, ports, protocol, length, TCP flags/seq/ack, ICMP type/id/seq, DNS id). Every later module works on any source.
- **Pipeline** (netmind/pipeline.py) runs the modules in order and saves each analysis as a **run**.
- **Interfaces:** command line (python -m netmind ...), Streamlit dashboard, FastAPI REST API (/health, /runs, /ask, /path, /routing ...).
- **Stack:** Python, Scapy, pandas, NumPy, scikit-learn, NetworkX, SQLite, Streamlit, Plotly, FastAPI, NS2.

5. Modules: what each does and how

Module	Folder	What it does and how
M1 Input	`netmind/capture/`	Scapy live capture (sniff, needs Npcap and admin) and pcap files (rdpcap); NS2 wired trace parser (event time from to type size flags fid src dst seq id); synthetic demo: 3 clients browsing (DNS, TCP handshake, HTTP/HTTPS, FIN) and pinging the gateway, with congestion 70-100 s, port scan 120-128 s from 192.168.1.66 and ping flood 150-158 s injected. Deterministic (seed 7), labelled synthetic.
M2 Traffic	`netmind/analyzer/`	Protocol from the IP header (TCP/UDP/ICMP) and well-known ports (53 DNS, 80 HTTP, 443 HTTPS). **Flow** = 5-tuple (src IP, src port, dst IP, dst port, protocol), both directions merged. Outputs protocol shares, flows, top talkers.
M4 Metrics	`netmind/performance/`	Per 2 s window: **latency** = SYN to SYN-ACK (ack = seq+1), ICMP echo request to reply (same id, seq), DNS query to response (same id), averaged per path; **throughput** = bytes × 8 / window; **loss** = (TCP retransmissions + unanswered pings) / (TCP data segments + pings); **jitter** = mean $\lvert RTT_i - RTT_{i-1} \rvert$ on the same path; packets/s; utilisation.
M5 Topology	`netmind/topology/`	NetworkX graph of hosts that exchanged packets; weights = median RTT or hops. **Dijkstra** shortest path. **Link State** (OSPF idea): each node knows the map, runs Dijkstra. **Distance Vector** (RIP idea, Bellman-Ford): neighbours exchange tables each round until nothing changes; rounds are reported. LS and DV give identical costs (tested); DV reroutes after a link failure.
M6 Storage	`netmind/storage/`	SQLite (data/netmind.db): runs table plus per-run metrics, anomalies, incidents, nodes, links, flows, protocols, RTT samples.
M3 Anomaly	`netmind/anomaly/`	10 features per window (packets/s, avg packet size, connections, connection duration, distinct destination ports, TCP/UDP ratio, bytes/s, latency, loss, retransmissions). Baseline = first 60 s of the demo (default 30%, at least 8 windows = 16 s; less → detection skipped, never "normal"). **Isolation Forest** (200 trees), score on 0-1 scale (≈0.5 normal, near 1 unusual); threshold = 99th percentile of baseline + 0.03 + 0.25/√n. **Deviation check**: robust $z = (x - \text{median}) / (1.4826 \times \text{MAD})$. Flag if $z \geq 6$, or forest score > threshold and $z \geq 4$. Consecutive flags → **incidents**, typed as congestion, port scan, flood, traffic spike, reroute.
M7 RAG	`netmind/rag/`	9 networking documents written for the project → 36 chunks (by section, ~140 words). TF-IDF (1-2 grams) by default, works offline; Sentence Transformers optional; cosine search (FAISS optional); top 4 chunks go into the prompt.
M8 Copilot	`netmind/copilot/`	Prompt = question + network state + incidents with evidence + retrieved knowledge [1]-[4]. Backends: offline rule-based (default, deterministic), Claude API, local Ollama; falls back to offline if the LLM is unreachable. Cites sources. Advisory only.
M9 Dashboard	`netmind/dashboard/`	Streamlit, 8 pages (section 8), light and dark themes; also the CLI and the REST API.
NS2	`ns2/`	netmind_scenario.tcl, trace parser (parse_trace.py), standard-library checker (check_trace.py), lab guide (docs/NS2_LAB_EXECUTION.md). Details in section 7.

6. Results to quote

Synthetic demo (python -m netmind demo, 180 s, seed 7: generated traffic, not a capture)

Injected event	Time	Detected as	Evidence
Congestion	70-100 s	Network congestion	Latency 146 ms vs 26.5 ms normal (5.5×); loss about 9.5% vs 0
Port scan	120-128 s	Possible port scan, from 192.168.1.66	311 destination ports per window vs about 10
Ping flood	150-158 s	Traffic flood, ICMP from 192.168.1.66	551 packets/s vs 41; average packet 104 vs 660 bytes

- 13,511 packets, 9 hosts, 1,731 flows; 90 windows, 23 flagged (all inside the three events).
- All three events found, with the right type, time and source.
- **False alarms** on normal-only synthetic traffic (5 lengths × 10 seeds, about 1,350 windows): 3 windows flagged, under 0.5%. Before the tuning it was 18.

Tests

- Automated pytest suite: capture, metrics, topology (LS = DV, DV reroutes), anomaly detection, NS2 parser, pipeline, RAG, Copilot, CLI, edge cases (empty and very short captures), dashboard logic and every dashboard page (Streamlit AppTest).
- On the laptop before the dashboard redesign: **61 passed, 2 skipped**. After git pull, run it again and quote the new number.

7. NS2 simulation

Status	State
Implemented	Done: TCL scenario, trace parser, checker reviewed and fixed
Validated	Done: parser and NetMind pipeline tested on a sample trace
Executed	**Done on 15 Oct 2026**: scenario run in NS2 on Ubuntu 24.04 (on the laptop); all 5 checks passed on the real trace

Scenario (5 nodes)

Time	Event	Expected
0.5 s	TCP (FTP) n0 → n3 starts, window 6 packets	No drops, normal delay
3.0-6.0 s	UDP CBR 0.9 Mb/s n1 → n3 on the 1 Mb n2-n3 bottleneck (queue 10)	Queue fills: drops, delay rises (congestion)
7.0 s	Link n2-n3 goes down	Distance Vector reroutes TCP via n4
8.5 s	Link n2-n3 back up	TCP returns to n2-n3
10 s	End	netmind.tr (trace) and netmind.nam (animation)

Real NS2 results (check_trace.py on the real trace)

Check	Result
No drops before the UDP burst (0.5-3 s)	PASS: 0 drops
Drops during the UDP burst (3-6 s)	PASS: 21 drops
Delay higher during the burst	PASS: 43.0 ms → 103.1 ms
Traffic rerouted via n4 during the link failure (7-8.5 s)	PASS
TCP delay higher on the backup path	PASS: 115.2 ms

- **Definitions** used for NS2: delay = receive time − send time (one-way, NS2 knows both); loss = dropped / (delivered + dropped), packets still in flight excluded; jitter per flow; a retransmission = same flow id and sequence number sent again.
- **Why NS2:** problems at known times in a repeatable network, so we can check NetMind's measurements and detection against the truth, and see Distance Vector rerouting.
- **Show:** the check output, NAM at 3 s / 7 s / 8.5 s, and the trace analysed in the dashboard (Incidents: congestion and link failure; Topology: fail n2-n3).

8. Dashboard pages

Page	What to point out
Overview	Whole run as one traffic chart with incidents as red bands; verdict ("3 incidents in 180 s"); six headline numbers with trend lines; latency and loss vs normal; protocols; top talkers; Copilot panel
Incidents	Anomaly score over time; per incident: severity, what stood out vs normal, zoomed chart, "Ask the Copilot about this incident"
Traffic	Packets and bytes by protocol; searchable flow table (try 192.168.1.66)
Performance	Latency, jitter, loss, throughput, packets/s, retransmissions with the normal level; RTT by type
Topology & routing	Host graph, Dijkstra path, Link State vs Distance Vector table; **Simulate a link failure** shows which routes change and DV rounds
Copilot	Chat with suggested and per-incident questions, sources, and the exact prompt given to the model
Compare runs	Two runs side by side (demo vs normal-only traffic)
Data & export	Markdown report and CSV downloads, raw tables, baseline, delete a run

Sidebar: **Start a new run** (demo with seed and length, normal-only traffic, upload a pcap or NS2 trace, live capture). Light/dark: the three-dot menu at the top right → Settings.

9. Live demo script (about 5 minutes)

Time	Do	Say
0:00	Terminal: python -m netmind demo	Synthetic traffic with three problems injected at known times; labelled synthetic, so we know the truth.
0:30	Point at the window table and incidents	Every 2 s we measure rate, throughput, latency, jitter, loss. The detector learned normal from the first 60 s and found three incidents, each with evidence.
1:15	Dashboard Overview	Red bands are the incidents; worst latency is 5.5× normal.
1:45	Copilot panel: "Is anyone attacking my network?"	It names the scan, the attacker and what to check, and cites its sources. Advisory only.
2:30	Incidents page; Ask about the port scan	Severity, features that stood out vs normal, zoomed chart.
3:00	Topology & routing: fail a link	Dijkstra path; LS and DV agree; after the failure DV recomputes and the table shows what changed.
3:45	Ubuntu: check_trace.py output, NAM	Real NS2 run: congestion at 3 s, link failure at 7 s, reroute via n4. All five checks pass.
4:30	Dashboard: switch to the NS2 run	Same pipeline on the NS2 trace: congestion and link-failure incidents.

If something goes wrong

Problem	Do
Dashboard will not open	Use the terminal demo and python -m netmind ask "..."; same results
Port 8501 busy	python -m netmind dashboard --port 8502
NAM does not open	Show screenshots, or the NS2 run in the dashboard (Incidents, Topology)
Old run shown	Pick it in the sidebar Run box, or Start a new run → Generate and analyse
Asked for a real capture	python -m netmind capture --duration 60 in an Administrator terminal (Npcap); short captures report "not enough traffic" honestly

10. Viva questions and answers

Is this a networks project or an AI project?

Networks first: capture, protocol analysis, RTT/loss/jitter measurement, flows, topology, Dijkstra, Link State, Distance Vector and NS2 are the core. ML and the LLM only interpret what those measure.

How do you measure latency without sending anything?

From request/reply pairs already in the traffic: TCP SYN and SYN-ACK (ack = seq + 1), ICMP echo request and reply (same id and sequence), DNS query and response (same transaction id).

How do you know packets were lost if you never saw them?

Inferred: TCP resends lost data with the same sequence number, and a ping without a reply is lost. In NS2 the trace records drops directly (d events).

What is jitter, and why per path?

Variation in delay between consecutive packets. Different paths have different base delays; mixing them measures the difference between paths, not variation (it inflated jitter to about 20 ms).

What is a flow?

Packets sharing the 5-tuple (src IP, src port, dst IP, dst port, protocol); both directions merged.

Why Isolation Forest?

Unsupervised (no labelled attacks needed), fast; unusual windows are isolated in fewer random splits. Score = 2^(−average path length / c(n)): about 0.5 normal, near 1 unusual.

Why an extra deviation check?

The forest cannot split on a feature that never varied in training (loss is 0 in normal traffic), so it missed congestion. The robust z-score catches it and explains which features stood out.

How do you avoid false alarms?

$z \geq 4$ required behind a forest flag, a stricter threshold for short baselines, and a 16 s minimum baseline. Normal-only traffic: 3 of about 1,350 windows flagged (was 18).

Dijkstra vs Bellman-Ford?

Dijkstra takes the closest unfinished node each step and needs non-negative weights. Bellman-Ford relaxes all edges repeatedly, handles negative weights and is the basis of Distance Vector.

Link State vs Distance Vector?

LS (OSPF): routers flood link states, know the whole map and run Dijkstra. DV (RIP): routers only exchange distance tables with neighbours; simpler, slower to converge, can count to infinity.

What does NS2 add?

Problems at known times in a repeatable network, so NetMind's results can be checked against the truth, plus Distance Vector rerouting after a link failure.

What is RAG?

Retrieval-Augmented Generation: retrieve the most relevant knowledge chunks and add them to the prompt, so answers rest on correct theory and cite sources.

Why TF-IDF and not only embeddings?

Works offline with no model download; Sentence Transformers can be switched on.

What if the LLM is wrong?

Numbers come from the database, not the LLM; the prompt restricts it to the given data with citations; the Copilot only advises; offline mode is deterministic.

What about HTTPS traffic?

Payload is encrypted; we use headers, ports, sizes and timing only.

Why is the demo synthetic?

So the truth is known: we inject problems at known times and check NetMind finds them. Real captures and NS2 traces go through exactly the same pipeline.

Ethics?

Only capture networks you own or may monitor; the Copilot never changes the network.

11. Limitations and next steps

- Demo data is synthetic; thresholds were tuned on it and should be checked on real captures.
- Topology is logical (who talked to whom), not physical cabling.
- Passive loss is an estimate (retransmissions and unanswered pings).
- **Next:** analyse real campus/home captures; more NS2 scenarios (more nodes, other queue types); evaluate the Copilot answers; record the NS2 results and NAM screenshots in ns2/results.

12. Changes since Review 1

- NS2 Simulation Module added, as the faculty suggested; now executed in NS2 with all checks passing.
- Full implementation of all 9 modules, CLI, REST API and dashboard; automated tests.
- Fewer false alarms; honest "not enough traffic" instead of "no anomalies" for short captures.
- Redesigned dashboard with 8 pages, Copilot panel, link-failure simulator, light and dark themes.